

1938

A graduate student at MIT, Claude Shannon, combined the binary system (which was first created by... Leibniz!) with Boolean algebra and published a paper on the implementation of symbolic logic using relays. It is hard at first glance to understand the importance of this.

Let us spell it out: with the new, proposed system, it would be possible to design digital computers!

It is beyond our scope here to explain symbolic logic; we'll just say the little that follows. Suppose you were to assign symbols to facts, such as, you assign "A" to "This is Digit", and "B" to "This is a newspaper". Then, both A *and* B being true would be " $A \wedge B$ "; either A or B being true would be " $A \vee B$ ". In this particular case, " $A \wedge B$ " is False, and the latter is True. Now note that this can be captured by Boolean algebra, as we explained it. And if *that* can be married to the binary system (here, relays)—then we have a "killer app" as we call it these days: logic over the wires! Logic and reasoning in transistors and chips and what-not! Which means—*machines can now act on information!*



Claude Shannon married bits to logic

To borrow from thocp.net, "In his paper, which was widely circulated, Shannon showed how Boole's concepts of TRUE and FALSE could be used to represent the functions of switches in electronic circuits. It is difficult to convey just how important this concept was; suffice it to say that Shannon had provided electronics engineers with the mathematical tool they needed to design digital electronic circuits, and these techniques remain the cornerstone of digital electronic design to this day."